

Impact of Technology: Technical Worksheet

Grade 8 • Worksheet • About 25 minutes

Work with pen and paper. Use color pencils if you like. You do not need a computer or the internet.

Name: _____

Grade/Section: _____

Date: _____

How to use this worksheet

- Do the parts **in order** from top to bottom.
- Each part has a **time** so you know how long to spend.
- Read every instruction before you write. You can do this on your own.

Part 1 – What you will learn (about 3 minutes)

What you will learn: Technology impact means the effects a tool or system has on people, places, or other systems. Some effects are **intended** (planned) and some are **unintended** (not planned). In this worksheet you will learn to notice **systems** (inputs → process → outputs), classify impacts, and think about **trade-offs** when choosing technology.

Do this:

1. **In one sentence**, what is technology impact?

2. **Name one technology** you use every day (e.g. phone, laptop, game console).

Part 2 – Intended and unintended impact (about 5 minutes)

Intended impact is what designers or leaders hope will happen (e.g. faster work, easier communication). **Unintended impact** is what happens anyway, even if no one planned it (e.g. some people are left out, new problems appear). **Efficiency** means doing more with less time or effort. **Automation** means machines or software do tasks that people used to do.

Real-life scenario: A school switches to online assignment submission. Teachers grade faster and papers are not lost. Some students do not have reliable internet at home and cannot upload work on time.

Discuss with a partner or reflect in writing: What was the **intended** impact? What might be **unintended**?

Do this:

1. **List 2 intended effects** and **2 unintended effects** of online submission.

Intended:

1.

2.

Unintended:

1.

2.

2. **Which problem should the school try to fix first?** Write 1, 2, 3, or 4 next to your top priority. **In one sentence**, why?

Priority: _____

Reason:

Part 3 – Data trails in digital systems (about 5 minutes)

Many apps and online services leave **data trails**—records of what you do (logs, clicks, time spent, location). Data is not good or bad by itself; it depends on **how** it is collected, stored, and used.

Real-life scenario: A fitness app tracks your steps and your location. The company wants to sell **aggregated** data (combined information from many users) to advertisers.

Discuss with a partner or reflect in writing: What kinds of data might this app collect? Who could benefit? Who could be harmed?

Do this: Match each data type to why it could matter. Use the word bank. Each word is used once.

Word bank: *Location · Screen time · Contacts · Purchase history*

Data type	Why it could matter (your answer)
Where you walk or run	_____
How long you use the app each day	_____
Names of people in your phone list	_____
Things you bought through linked stores	_____

Part 4 – Resource and environmental impact (about 5 minutes)

Technology also has **resource impact**: it uses **energy**, needs materials to build devices, and creates **e-waste** (old electronics thrown away). A longer **device lifespan** (using a device longer before replacing it) often means less waste.

Scenario: A school can **Option A:** buy new tablets every 2 years, or **Option B:** repair older tablets and buy fewer new ones.

Do this:

	Option A (new every 2 years)	Option B (repair and reuse)
Cost (high / medium / low)	_____	_____
E-waste (high / medium / low)	_____	_____
Learning disruption (high / medium / low)	_____	_____

1. **Circle** the option you think is better overall: **A** or **B**. **In one sentence**, explain your choice.

Choice: _____

Reason:

2. **Critical thinking:** Can a choice that is better for the school still harm someone else (e.g. workers, communities, the environment)? **In one or two sentences**, give an example.

Part 5 – Evaluating trade-offs (about 5 minutes)

Choosing technology often means **trade-offs**: you gain something and give up something else. More features can mean more **data collection** or more **energy use**.

Discuss with a partner or reflect in writing: Why might a “free” app still have a cost for users?

Do this: A school needs a chat app for class announcements.

- **Option A:** Free app with ads; collects usage data to show targeted ads.
- **Option B:** Paid app with no ads; fewer features; limited storage.

1. **Which option would you recommend?** Circle **A** or **B**. **In one or two sentences**, justify your choice.

Choice: _____

2. **Write one question** you would ask before the school adopts either app (think about privacy, cost, or reliability).

Part 6 – Check what you learned (about 2 minutes)

Checklist. Put a ✓ in the box when it is true for you.

- ☐ I know what technology impact means (intended and unintended effects).
- ☐ I understand how efficiency and automation can change work and access.
- ☐ I understand data trails and why data types matter.
- ☐ I understand resource and environmental impact (energy, e-waste, lifespan).
- ☐ I have thought about trade-offs when choosing technology.

When all boxes are checked, you have reviewed the main ideas from this worksheet.